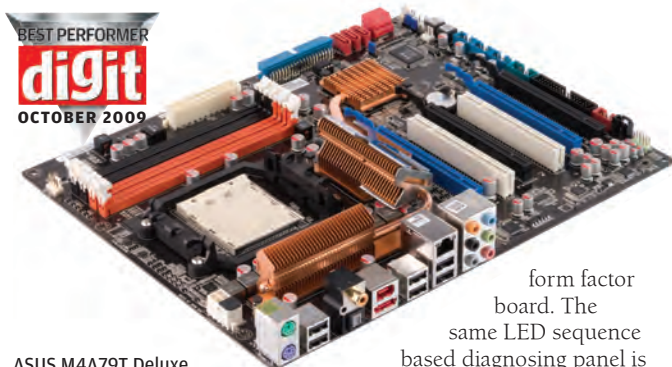


GT300

NVIDIA rumoured to follow AMD in displaying their GT300 GPU

LED displays

BenQ and NEC start producing LED powered displays



ASUS M4A79T Deluxe

these are just a little more expensive but come with an abundance of features. One of the first things you notice is that it's a completely solid-state capacitor board. There is a colourful LED powered display at the bottom of the board. There's a plastic sticker which in place to spread the light through the gap in the board. Perhaps, this could have been implemented a little better. A switch card is provided to make sure that the board uses the entire PCIe x16 bandwidth. The heat sink itself is very attractive. The heatsinks are connected by a copper heat pipe. The pipe is connected using a set of multiple clips within the heat sink so the heat exchange isn't as much as it could have been if the heat pipe were completely consumed by a block of heat sink for example. The two PCIe x1 slots are placed right below the first x16 slot. Although, the first one might not be usable, the other one will. There are also two PCI slots below the PCIe x16 slot. This board in terms of features is a winner for its price. All the three display connectivity options - VGA, DVI and HDMI are available. There's also an SATA port meant for eSATA placed right near the back of the board right next to the external eSATA port. There are buttons of all the power controls and also CMOS clearing. With everything kept in mind, this is a very good board to consider.

The TA790GX A3+ from Biostar is pretty similar in design to the TA785GE motherboard and is a full-fledged ATX board instead of a HTPC-friendly mATX

form factor board. The same LED sequence based diagnosing panel is present on this board. Like the TA785GE, this too has all three display connections - DVI, VGA and HDMI. The SATA ports are isolated and aren't hugging the side of the motherboard like on almost every other board. If you think of using L-shaped SATA connectors on the motherboard end, it won't be possible. It's kind of funny to see so many features printed on the motherboard. Everything from the kind of slots to every single connectivity such as the LAN, the 7.1 audio, the DVI ports, the Firewire connectors are printed in large characters on the motherboard. The power buttons and the CMOS clear jumper are positioned in one corner of the board.

The ECS A790GXM-A is another that uses the 790GX chipset. It's not really geared as a high-performance enthusiast board. The board is rather decent looking with its colourful slots and



Gigabyte MA790FX-UD5P

a dark PCB. The positioning of the Northbridge chip is unconventional and slightly towards the back-end of the board. The onboard Qimonda

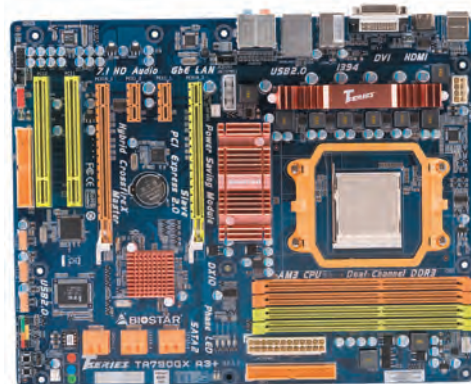
branded memory chip lies in between it and the processor. In terms of just features, it matches some of the best boards - in the BIOS and overclocking department, not so much. It's very basic BIOS. Amongst the two LAN ports and HDMI port, lies an ancient COMM port. The board isn't a fully solid state capacitor board.

The Gigabyte MA790FX-UD5P is a high-end board based on the AMD 790FX chipset. The 790FX is pretty much the 790GX without the on-board graphics solution. Cooling is similar to other boards except for a few things here and there. For example, there's a tiny isolated heat sink for the SATA chip. The package contains back panel for eSATA come with molex to SATA power connectors and eSATA cables as well. This should technically let you use any drive as an external drive very easily. The space between the two PCIe x16 slots is sufficient and there are three PCIe slots. The topmost one lies in the way of the Northbridge heat sink but Gigabyte has cleverly cut out space in between so allow long cards to be installed.

Gigabyte's other AMD powered board the MA790GP-UD4H board has three PCIe x1 slots, one of them right next to the heat sink and almost unusable. A line of USB slots on the board are available behind the second PCIe X16 slot. The heatsinks are connected by two pipes and the blue Gigabyte SilentPipe.

The MSI 790GX-G65 comes with the necessary cooling setup for the board. The largest

heat sink is used on top of the MOSFETs. The board layout is good and there's a good mix of PCI and PCIe x1 slots on it. It comes packed with plenty of features as well. The dedicated buttons for the motherboard controls is also a nice thing to have. The board is designed for overclockers and the brochures that come with it state it. There's a switch to quickly change the FSB speed physically. A little more cooling would have been nicer on a board designed for overclocking. The package we received came with a signed



Biostar TA790GX A3+

certificate of some benchmarks by those who run it. This is a slightly different treatment to a board that we've seen. It's nice to know that you've been sold a well tested product.

The ASUS M4A79T Deluxe is an AMD 790FX chipset board. Like the P45 boards from ASUS, this too uses the ADI2000B chip for the sound. The board doesn't come with any PCIe x1 or x4 sized slots. There are four PCIe x16 slots which can be utilized for any other cards. The Southbridge heat sink is rather plain but made of copper. ASUS being themselves have kept capacitors out of the way of any PCI and PCIe slots. The package contents are very basic. SATA cables and back panels are provided. No fancy fans or cables here.

Performance

The boards we tested scored almost identical performance figures across all chipsets. Obviously, a lot depended on